United States Patent Application Publication	20250269107
Kind Code	A1
Publication Date	August 28, 2025
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System for Infusion of Hazardous Drugs

Abstract

An infusion system includes an infusion adapter having a spike configured to pierce a seal of an infusion container, a grip portion extending radially outward relative to the spike, and a connection interface having a membrane, with the infusion adapter defining a passageway extending from a first end of the infusion adapter to the second end of the infusion adapter and the membrane sealing the passageway at the second end of the infusion adapter, and a connection adapter having a spike port and a connector body including a luer connector.

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Family ID: **1000007845384**

Appl. No.: **18/584035**

Filed: **February 22, 2024**

Publication Classification

Int. Cl.: **A61M5/14** (20060101); **A61M5/142** (20060101); **A61M39/10** (20060101)

U.S. Cl.:

CPC **A61M5/1407** (20130101); **A61M5/1424** (20130101); **A61M39/10** (20130101); A61M2039/1077 (20130101); A61M2039/1088 (20130101)

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Disclosure

[0001] The present disclosure relates generally to a system for the infusion of hazardous drugs, including an infusion adapter and a connection adapter.

Description of the Related Art

[0002] Reconstituting, transporting, and administering hazardous drugs, such as cancer treatments, can put health care providers at risk of exposure to these medications and present a major hazard in the health care environment. For example, nurses treating cancer patients risk being exposed to chemotherapy drugs and their toxic effects when administering these potentially harmful drugs due to insecure connections or inadvertent disconnection of the delivery devices or adapters used to transport or administer such drugs. Unintentional chemotherapy exposure can affect the nervous system, impair the reproductive system, and bring an increased risk of developing blood cancers in the future. In order to reduce the risk of health care providers being exposed to toxic drugs, the secure, closed transfer of these drugs is important.

SUMMARY OF THE INVENTION

[0003] In one aspect or embodiment, an infusion system includes an infusion adapter having a spike configured to pierce a seal of an infusion container, a grip portion extending radially outward relative to the spike, and a connection interface having a membrane, with the infusion adapter defining a passageway extending from a first end of the infusion adapter to the second end of the infusion adapter and the membrane sealing the passageway at the second end of the infusion adapter. The system further includes a connection adapter having a spike port and a connector body comprising a luer connector.

[0004] The spike port may be connected to the connector body via an interference fit. The luer connector of the connector body may include a male luer lock connector. The grip portion may be annular. The grip portion may be bowl-shaped. A maximum diameter of the grip portion may be 17 mm. The grip portion may include first and second planar surfaces. The grip portion may include first and second openings.

[0005] The system may further include a syringe adapter configured to receive the connection interface of the infusion adapter and configured to be connected to the connection adapter via the luer connector. The connection interface may further include a plurality of cantilever arms each having a protrusion.

[0006] In a further aspect or embodiment, a method of using the infusion system described above includes: providing a plurality of infusion containers; inserting the spike of the infusion adapter within a port of a first infusion container of the plurality of infusion containers; connecting the connection adapter to the syringe adapter via the luer connector; and connecting the syringe adapter to the infusion adapter via the connection interface.

[0007] The method may further include placing the spike port of the connection adapter in fluid communication with a patient infusion line. The method may further include activating an infusion pump. Each of the plurality of infusion containers may include a flexible intravenous bag.

[0008] The method may further include: disconnecting the first infusion container from the syringe adapter; inserting a spike of a second infusion adapter within a portion of a second infusion container of the plurality of infusion containers; and connecting the syringe adapter to the second infusion adapter via the connection interface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A more complete understanding of the present disclosure, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description

when considered in conjunction with the accompanying drawings, wherein:
[0010] FIG. **1** is a schematic view of a conventional primary infusion arrangement;
[0011] FIG. **2** is a schematic view of a conventional secondary infusion arrangement;
[0012] FIG. **3** is a schematic view of a conventional multiway set infusion arrangement;
[0013] FIG. **4** is a perspective view of an infusion system according to one aspect or embodiment of the present application;
[0014] FIG. **5** is perspective view of a conventional syringe adapter;
[0015] FIG. **6** is a perspective view of an infusion adapter of the system of FIG. **4** according to one aspect or embodiment of the present application;
[0016] FIG. **7** is a front view of a connection adapter of the system of FIG. **4** according to one aspect or embodiment of the present application;
[0017] FIG. **8** is an exploded view of the connection adapter of FIG. **7**;
[0018] FIG. **9** is a partial cross-sectional view of the connection adapter of FIG. **7**; and
[0019] FIG. **10** is a schematic view of the infusion system of FIG. **4**, showing the infusion system in use.
DETAILED DESCRIPTION OF THE INVENTION
[0020] The following description is provided to enable those skilled in the art to make and use the described aspects contemplated for carrying out the invention. Various modifications, equivalents, variations, and alternatives, however, will remain readily apparent to those skilled in the art. Any and all such modifications, variations, equivalents, and alternatives are intended to fall within the spirit and scope of the present invention.
[0021] As used herein, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.
[0022] Spatial or directional terms, such as “left”, “right”, “inner”, “outer”, “above”, “below”, and the like, relate to the embodiments or aspects as shown in the drawing figures and are not to be considered as limiting as the embodiments or aspects can assume various alternative orientations.
[0023] All numbers used in the specification and claims are to be understood as being modified in all instances by the term “about”. By “about” is meant within plus or minus twenty-five percent of the stated value. However, this should not be considered as limiting to any analysis of the values under the doctrine of equivalents.
[0024] Unless otherwise indicated, all ranges or ratios disclosed herein are to be understood to encompass the beginning and ending values and any and all subranges or subratios subsumed therein. For example, a stated range or ratio of “1 to 10” should be considered to include any and all subranges or subratios between (and inclusive of) the minimum value of 1 and the maximum value of 10; that is, all subranges or subratios beginning with a minimum value of 1 or more and ending with a maximum value of 10 or less. The ranges and/or ratios disclosed herein represent the average values over the specified range and/or ratio.
[0025] The terms “first”, “second”, and the like are not intended to refer to any particular order or chronology, but refer to different conditions, properties, or elements.
[0026] All documents referred to herein are “incorporated by reference” in their entirety.
[0027] The term “at least” is synonymous with “greater than or equal to.”
[0028] As used herein, “at least one of” is synonymous with “one or more of.” For example, the phrase “at least one of A, B, or C” means any one of A, B, or C, or any combination of any two or more of A, B, or C. For example, “at least one of A, B, and C” includes A alone; or B alone; or C alone; or A and B; or A and C; or B and C; or all of A, B, and C.
[0029] Referring to FIGS. **1-3**, conventional infusion arrangements are shown, including a primary infusion arrangement (FIG. **1**), a secondary infusion arrangement (FIG. **2**), and a multiway set infusion arrangement (FIG. **3**). As shown in FIG. **1**, with primary infusion, an intravenous bag **2** with saline is connected to a patient with an infusion pump **4** delivering a predetermined flow rate or volume of saline or other fluid to the patient. For delivery of chemotherapeutic medication, a

separate intravenous bag **6** with the chemotherapeutic medication is connected to a port **8** of the patient line via a syringe adapter and patient connector **10**, with a second infusion pump **12** delivering a predetermined flow rate, volume, or dose of chemotherapeutic medication to the patient. The separate intravenous bag **6** is accessed via an infusion adapter **14**. As shown in FIG. **2**, with secondary infusion, rather than providing the second infusion pump **12** and connecting downstream of the first infusion pump **4**, the separate intravenous bag **6** is connected upstream of the first infusion pump **4**. As shown in FIG. **3**, with a multiway set infusion, a multiway adapter **16** having a plurality of ports is connected to a pre-med intravenous bag **18**, a saline infusion bag **20**, and one or more chemotherapeutic infusion bags **22**, with infusion delivery via a single infusion pump **24**.

[0030] Referring to FIGS. **4-10**, in one aspect or embodiment of the present application, an infusion system **100** includes an infusion adapter **102** including a spike **104** configured to pierce a seal of an infusion container, a grip portion **106** extending radially outward relative to the spike **104**, and a connection interface **108** including a membrane **110**. The infusion adapter **102** defines a passageway **112** extending from a first end **114** of the infusion adapter **102** to the second end **116** of the infusion adapter **102**, with the membrane **110** sealing the passageway **112** at the second end **116** of the infusion adapter **102**. The infusion system **100** also includes a connection adapter **120** including a spike port **122** and a connector body **124** including a luer connector **126**. The infusion system **100** is configured to eliminate priming in pharmacy, decrease priming at bedside, eliminate uncontrolled spiking of intravenous bags, and reducing the number of intravenous lines utilized during infusion. The use of the infusion system **100** is discussed in more detail below.

[0031] Referring to FIGS. **4**, **7**, **8**, and **9**, the spike port **122** is connected to the connector body **124** via an interference fit with interlock features, although other suitable connection arrangements, such as a threaded arrangement, integrally forming the components, a luer connection, or adhering the components, may be utilized. The luer connector **126** of the connector body **124** is shown as a male luer lock connector, although other suitable connection arrangements, such as a threaded arrangement or friction fit connection, may be utilized. The connector body **124** defines at least one opening **128**, which is configured to visualize the spike port **122** when the spike port **122** is connected to the connector body **124**. An adapter passageway **129** extends through the connection adapter **120**, as shown in FIG. **9**.

[0032] Referring to FIGS. **4** and **6**, the grip portion **106** is annular, although other suitable shapes may be utilized. In some aspects or embodiments, the grip portion **106** has a maximum diameter of 17 mm, which reduces the possibility of interfering with other devices when attached to an infusion container and reduces the risk of inadvertent disconnection from the infusion container. In some aspects or embodiments, the grip portion **106** is bowl-shaped, which is configured to maximize the hand comfort of healthcare workers using the infusion adapter **102**. The grip portion **106** includes first and second planar surfaces **130**, **132**. The first and second planar surfaces **130**, **132** are configured to be engaged or grasped by a hand of a healthcare worker, and to enable consistent manufacturing assembly that minimizes errors such as misalignment between components. The grip portion **106** includes first and second openings **134**, **136**. The first and second openings **134**, **136** are aligned with the first and second planar surfaces **130**, **132** in a longitudinal direction of the infusion adapter **102**, although other suitable positions of the first and second planar surfaces **130**, **132** may be utilized. The first and second openings **134**, **136** are configured to allow visualization of a port of an intravenous bag to ensure the spike **104** of the infusion adapter **102** is sufficiently engaged with the intravenous bag. The spike **104** of the infusion adapter **102** includes an enlarged portion **138** positioned between the first and second ends **114**, **116** of the infusion adapter **102**. The enlarged portion **138** is configured to secure the infusion adapter **102** to the intravenous bag. The connection interface **108** of the infusion adapter **102** includes a plurality of cantilever arms **140** with a protrusion **142** that are configured to interact and connect to a corresponding connection interface of a mating component.

[0033] Referring to FIGS. **4** and **10**, in some aspects or embodiments, the infusion system **100** includes a syringe adapter **150** configured to receive the connection interface **108** of the infusion adapter **102** and configured to be connected to the connection adapter **120** via the luer connector **126**. An example of the syringe adapter **150** is shown in FIG. **5**. The syringe adapter **150** is configured to receive at least a portion of the connector interface **108** of the infusion adapter **102** within an opening **152**, and is further configured to connect to a syringe (not shown) or other medical device or fluid container via a connector **154**, such a female luer connection. The syringe adapter **150** is similar to and operates in a similar manner as the syringe adapter shown and described in U.S. Pat. No. 10,744,315, which is hereby incorporated by reference in its entirety. As shown in FIG. **4**, the threads of the connector **152** are covered by the luer connector **126** of the connector body **124** when the syringe adapter **150** is connected to the connector body **124**, which provides an indication that the syringe adapter **150** is securely connected to the connector body **124**.

[0034] Referring to FIG. **10**, a method **200** of using the infusion system **100** according to one aspect or embodiment of the present application includes: providing a plurality of infusion containers **202**; inserting the spike **104** of the infusion adapter **102** within a port of a first infusion container **204** of the plurality of infusion containers **202**; connecting the connection adapter **120** to the syringe adapter **150** via the luer connector **126**; and connecting the syringe adapter **150** to the infusion adapter **102** via the connection interface **108**.

[0035] Referring again to FIG. **10**, the method **200** may also include placing the spike port **122** of the connection adapter **120** in fluid communication with a patient infusion line **206**. The patient infusion line **206** may include a drip chamber **208** and an infusion pump **210**. The method **200** may include activating the infusion pump **210** to deliver a predetermined flow rate or volume of medication or fluid from one of the plurality of infusion containers **202**. As shown in FIG. **9**, each of the plurality of infusion containers **202** is a flexible intravenous bag, although other suitable containers may be utilized.

[0036] Referring again to FIG. **10**, in some aspects or embodiments, the method **200** further includes: disconnecting the first infusion container **204** from the syringe adapter **150**; inserting a spike **104** of a second infusion adapter **102** within a portion of a second infusion container **212** of the plurality of infusion containers **202**; and connecting the syringe adapter **150** to the second infusion adapter **102** via the connection interface **108**. The first infusion container **204** may include saline and the second infusion container **212** may include pre-meds or chemotherapeutic medication.

[0037] In some aspect or embodiments, the spike **104** of the infusion adapter **102** may be connected to one of the plurality of infusion containers **202** in a controlled environment, fume hood, and/or biosafety cabinet. The connection adapter **120** can be threaded onto the connector **154** of the syringe adapter **150** with the spike port **122** of the connection adapter **120** connected to the patient infusion line **206**, with the syringe adapter **150** connected to the connection interface **108** of the infusion adapter **102** to place contents of one of the infusion containers **202** in fluid communication with the patient infusion line **206**. The connection between the syringe adapter **150** and the connection interface **108** of the infusion adapter **102** ensures a closed transfer of fluid and prevents leakage or discharge of chemotherapeutic fluids from the infusion system **100**.

[0038] It will be appreciated by persons skilled in the art that the present disclosure is not limited to what has been particularly shown and described herein above. In addition, unless mention was made above to the contrary, it should be noted that all of the accompanying drawings are not to scale. Of note, the system components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure, so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Moreover, while certain embodiments or figures described herein may illustrate features not expressly indicated on other figures or embodiments, it is understood that the features and components of the examples disclosed herein are not necessarily exclusive of each other and may be included in a variety of different combinations or configurations without departing from the scope and spirit of the disclosure. A variety of modifications and variations are possible in light of the above teachings without departing from the scope and spirit of the disclosure, which is limited only by the following claims.

Claims

1. An infusion system comprising: an infusion adapter comprising a spike configured to pierce a seal of an infusion container, a grip portion extending radially outward relative to the spike, and a connection interface comprising a membrane, the infusion adapter defining a passageway extending from a first end of the infusion adapter to the second end of the infusion adapter, the membrane sealing the passageway at the second end of the infusion adapter; and a connection adapter comprising a spike port and a connector body comprising a luer connector.
2. The infusion system of claim 1, wherein the spike port is connected to the connector body via an interference fit.
3. The infusion system of claim 1, wherein the luer connector of the connector body comprises a male luer lock connector.
4. The infusion system of claim 1, wherein the grip portion is annular.
5. The infusion system of claim 1, wherein the grip portion is bowl-shaped.
6. The infusion system of claim 1, wherein a maximum diameter of the grip portion is 17 mm.
7. The infusion system of claim 1, wherein the grip portion comprises first and second planar surfaces.
8. The infusion system of claim 1, wherein the grip portion comprises first and second openings.
9. The infusion system of claim 1, further comprising a syringe adapter configured to receive the connection interface of the infusion adapter and configured to be connected to the connection adapter via the luer connector.
10. The infusion system of claim 1, wherein the connection interface comprises a plurality of cantilever arms each having a protrusion.
11. A method of using an infusion system, the method comprising: providing a plurality of infusion containers, each of the plurality of infusion containers including a seal; providing a syringe adapter; providing an infusion adapter, comprising a spike configured to pierce a seal of the plurality of infusion containers, the infusion adapter also including a grip portion extending radially outward relative to the spike, and a connection interface comprising a membrane, the infusion adapter defining a passageway extending from a first end of the infusion adapter to the second end of the infusion adapter, the membrane sealing the passageway at the second end of the infusion adapter; and a connection adapter comprising a spike port and a connector body comprising a luer connector; inserting the spike of the infusion adapter within the seal of a first infusion container of the plurality of infusion containers; connecting the connection adapter to the syringe adapter via the luer connector; and connecting the syringe adapter to the infusion adapter via the connection interface.
12. The method of claim 11, further comprising: placing the spike port of the connection adapter in fluid communication with a patient infusion line.
13. The method of claim 12, further comprising: activating an infusion pump.
14. The method of claim 11, wherein each of the plurality of infusion containers comprises a flexible intravenous bag.
15. The method of claim 11, further comprising: disconnecting the first infusion container from the syringe adapter; inserting a spike of a second infusion adapter within a portion of a second infusion

container of the plurality of infusion containers; and connecting the syringe adapter to the second infusion adapter via the connection interface.